AI Compiler Performance Engineering

Auto-Loops Books — ch02

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Chapter 1

Two Kernel Design Mindsets: Operator-Driven vs Dataflow-Driven

Chapter **??** showed *what* breaks in decode: launch storms, HBM round trips, and hardware-specific bottlenecks (Davies *et al.*, 2025). A durable industry misconception is that the fix is "one more fused op" inside an otherwise operator-first stack—when the schedule itself still exports every intermediate to global memory between mathematically adjacent compute stages (Wang, 2026). This chapter completes the pivot: from **operator-driven** schedules (compute-first, memory-second) to **dataflow-driven** schedules (residency-first, compute as a consequence of placement) (Sun, 2019). Readers should leave with a decision procedure: given a decoder bottleneck measurement (kernels/token, bytes/token, HDBI, R_{floor}), classify whether the next engineering dollar belongs in op micro-tuning or dataflow replanning—and default to replanning when Chapter 1 diagnostics show orchestration or HBM export dominance (Kim, 2025).

1.1 Operator First

In production, most CUDA teams still begin decoder optimization by listing operators: LayerNorm, QKV projection, attention, MLP, residual adds (Dao *et al.*, 2023). Each op receives a tuned kernel; the runtime stitches them with separate launches (Liu *et al.*, 2024). The failure mode is predictable: *compute* micro-optimizations improve individual kernels while end-to-end ms/token stalls because



Figure 1.1: Operator-driven schedule: each op reads/writes global memory.

every boundary writes activations to HBM and pays another launch (NVIDIA, 2022). Operator-first design treats memory as an implementation detail behind each op (AMD, 2024a). PyTorch eager mode is the canonical embodiment: the graph is correct, but the schedule is emergent (SemiAnalysis, 2024). FlashAttention grafts faster replacements for single nodes; it does not change export/import between nodes unless a compiler pass merges regions (Dao *et al.*, 2022). On GPUs this yields hundreds to thousands of kernels per decode token; on CPUs LLC churn; on NPUs illegal dynamic schedules (Goodfellow *et al.*, 2016). Teams can spend quarters shaving nanoseconds inside MMA loops while HDBI remains near zero. The operator-first workflow also fragments accountability: kernel owners optimize "their" op, while no team owns the HBM edges between ops—yet those edges dominate decode. Compiler stacks that expose only per-op autotuning (template GEMMs, library dispatch tables) reinforce this silo unless a higher pass owns cross-op lifetime analysis. Dataflow-first teams invert the org chart: one schedule owner, many op specializations *inside* fused regions (Zhao, 2025).

Example 1.1.1. A dense 7B decoder layer on H100 at batch-1 may spend ∼15–25 μs in device compute per op group but ∼30–50 μs waiting on launches and framework translation between groups. An operator-first team that speeds each compute group by 20% saves only ∼3–5 μs aggregate while launch gaps remain. A dataflow-first team that fuses three groups into one dispatch may eliminate two launch gaps (∼60–100 μs class savings on launch-bound configs) even if per-op compute is unchanged. The example is qualitative—exact numbers vary by model and driver—but the *ranking* of leverage is what production teams must internalize.

1.2 Data Residency

The root cause is not insufficient FLOPs; it is **data residency** violated at every layer boundary. Activations that could live in registers or shared memory for the full QKV→attention→MLP chain are materialized to global memory after each sub-step. Each round trip pays full HBM bandwidth and pollutes CPU caches. Residency is a compile-time allocation problem: which tensors coexist, which tier holds them, for how many fused stages? On GPUs the tiers are registers → shared → L2 → HBM; on CPUs cache and NUMA; on XDNA/AIE hardware-enforced tile

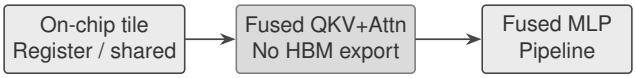


Figure 1.2: Dataflow-driven schedule: activations stay on-chip across fused stages.

buffers (AMD, 2024b). When residency succeeds, arithmetic intensity rises because bytes moved drop even if FLOPs stay constant. PagedAttention improves KV capacity but does not guarantee on-chip residency inside a fused decoder region (Kwon *et al.*, 2023). Quantization changes residency economics: int4 weights reduce bytes streamed per layer, but only when kernels truly consume compressed layouts without dequant scratch in HBM. A residency plan must therefore co-design precision, layout, and fusion: nf4 weights with bf16 activations still ping-pong if every op dequantizes to global buffers. Industrial teams should measure *bytes per token* and *unique HBM touch count* per layer alongside FLOPs—metrics that residency planning directly minimizes.

Residency analysis also exposes **anti-locality** in popular serving features. Continuous batching improves throughput by reusing weight matrices across requests, but if each request's activations still round-trip through HBM inside the layer, tail latency per request remains decode-bound. Speculative decoding adds draft-model traffic—another residency consumer—that operator-first stacks often bolt on without revisiting fusion boundaries. Dataflow planners must account for multi-model residency simultaneously: base model weights, draft model weights, and KV streams competing for bandwidth and on-chip space. The book's later compiler passes treat these as first-class buffer interference constraints rather than afterthought flags.

1.3 Dataflow Philosophy

Dataflow design names the schedule before operators. Four commitments recur across targets:

1. **Static placement** where possible—tensor lifetimes planned before codegen; NPUs require static ADF graphs.

2. **On-chip residency first**—fusion depth keeps producer–consumer pairs in the fastest tier.

3. **Pipelined movement**—TMA on Hopper and DMA on AIE hide transfers when the graph exposes prefetch edges.

Table 1.1: Design mindset vs. hardware enforcement model.

Target	Data residency model
CUDA GPU	Software-managed hierarchy
XDNA / AIE	Hardware-enforced spatial buffers
CPU	Cache hierarchy + explicit blocking

4. **Minimal synchronization**—dispatch unit is the pipeline stage, not each elementary op.

Porting a CUDA operator list to XDNA without a static pipeline yields no legal schedule. On GPUs, ignoring dataflow still "works"; on NPUs compile fails—explaining why operator-first CUDA expertise does not automatically transfer to edge fleets. Static dataflow does not mean frozen models: it means shapes and buffer slots are known at compile boundaries (per deployment SKU), with guarded recompilation when service configs change. Serving systems that mix dynamic batching with static fusion must separate *scheduling* concerns (which requests share a batch) from *kernel* concerns (what stays on-chip during a fused step). Confusing the two produces graphs that are dynamically batched at the framework level but still operator-fragmented inside each batch member—the worst of both worlds for tail latency.

Dataflow philosophy also reframes **correctness work**. Operator-first testing validates per-op numerical parity against reference implementations. Dataflow-first testing must add **schedule parity**: fused pipelines must match unfused references within tolerance while preserving tier assignments across backends. Regression suites therefore include golden activation dumps at stage boundaries during development, then strip those dumps in production builds once parity is proven—a pattern YiRage CI adopts for multi-target codegen. The extra testing cost pays back when a single fusion bug would otherwise ship as a 1.5× silent slowdown on only one hardware class.

1.4 Cross Hardware

Cross-hardware comparison is not an appendix exercise—it is how teams avoid shipping GPU-tuned schedules that silently fail on edge NPUs or underperform on CPU inference nodes co-located with GPU fleets. The book's benchmark harness (Appendix B) always reports at least three SKUs when hardware access allows: one bandwidth-rich GPU, one orchestration-sensitive GPU, and one CPU or NPU

class target. Mindset differences show up as *different optimal fusion depths* on the same mathematical model, not as different models.

For **CUDA GPUs**, shared memory sizing, register pressure, and occupancy set fusion depth. CUDA Graphs amortize launches but replay a fixed dataflow; they do not invent residency that eager boundaries destroyed (NVIDIA, 2024). Hopper TMA raises async movement ceilings but needs static copy edges. For **CPUs**, residency maps to **cache** blocking and NUMA placement; GPU-style mega-fusion can evict KV working sets. SIMD kernels benefit when dataflow keeps head layouts hot across QKV and MLP epilogues. For **XDNA / AIE**, residency is hardware-enforced: spatial tiles, DMA routes, compile-time buffer tables. Dynamic shapes and per-op dispatch are mismatches; the dataflow graph *is* the program. One mindset expresses differently per target, but operator-first lists fail everywhere for the same reason—no global residency contract. Table 1.1 summarizes enforcement models; the engineering consequence is that **fusion granularity** differs by class: GPUs tolerate medium fusion with shared-memory staging; CPUs need cache-aware blocking; NPUs require layer-scale or multi-layer static regions. Cross-hardware portability is portability of the *dataflow graph and lifetime annotations*, not of individual op implementations. YiRage backends swap codegen while preserving tier assignments—the topic of Parts IV–VI.

1.5 Iron Rules

Rule 1 — Registers hold ephemeral scalars and narrow vectors, not whole feature maps; spills mean the split is wrong. **Rule 2 — Shared memory / LDS / on-chip SRAM** holds fusion regions; depth ends at capacity conflicts, not when the op list ends. **Rule 3 — Global memory** is for weights, KV streams, and true outputs—not ping-pong activations between adjacent ops. Violations produce Chapter 1 pathologies: high kernels/token and low R_{floor} when export/import dominates. Use iron rules as release gates: if a proposed schedule inserts a global round trip between ops that share a producer–consumer edge in the mathematical graph, reject it unless profiling proves the tier overflow is unavoidable. On GPUs, "unavoidable" should trigger shared-memory tiling or pipeline splitting, not acceptance of HBM ping-pong. On NPUs, overflow should fail compile and force graph refactoring—not runtime fallback loops on the host.

Register pressure on GPUs frequently triggers the false comfort of "spilling is fine because the kernel still runs." In decode, spilled activations often become HBM traffic that negates fusion. Iron Rule 1 therefore pairs with occupancy discipline: if widening tile sizes forces spills, shrink fusion depth instead of celebrating higher

theoretical occupancy on paper. CPU analogs apply to stack spills and false sharing when fusing across threads without cache-line-aware partitioning.

1.6 MegaKernel Preview

A **MegaKernel** dispatches an entire decoder layer with activations resident across QKV, attention, MLP, norms, and residuals—the antithesis of Figure 1.1. MegaKernels are fusion schedules with staged pipelines, not one giant matmul. Chaining vendor libraries without shared residency drops launches modestly but often not HBM traffic. True MegaKernels collapse both; TaxBreak still shows order-of-magnitude launch gaps on MoE graphs after partial fusion. They trade flexibility for performance—acceptable when model revisions are pinned per fleet. Part V develops manual craft; Part VI automates the same decisions in YiRage. MegaKernel engineering also clarifies **where not to fuse**: attention kernels that require global KV streaming may remain distinct stages while MLP+norm+residual fuse around them—dataflow does not mandate monolithic single-kernel binaries, it mandates explicit staging with minimal exports. Flash-Decoding parallelizes KV dimension while keeping decode batch-1 viable; a MegaKernel schedule must *compose* with such attention stages rather than naively inlining them and destroying register budgets.

1.7 Automation Path

YiRage ingests decoder graphs, plans lifetimes globally, runs fusion passes, and emits CUDA, CPU, or XDNA backends from one dataflow IR (Chen, 2026). The automation path encodes expert workflow—probe hardware, plan residency, fuse, codegen, regress—as compiler logic. Each backend re-instantiates the same dataflow with different tier rules: GPU shared-memory fusion and graph capture; CPU cache blocking; XDNA static buffers and DMA ping-pong. YiRage fails compile when residency cannot be proved—preferable to silent production slowdowns. Optimize dataflow first; let operators fall out of the schedule second. Automation must preserve human-debuggable artifacts: lifetime graphs, fusion decision logs, and regression tables comparing bytes/token and kernels/token against PyTorch eager and framework servers. Without those artifacts, compiler automation reintroduces opacity—teams cannot tell whether a regression is a bad tile size or a violated iron rule. YiRage treats diagnostics as first-class outputs alongside generated kernels, echoing TaxBreak and memory-floor methodologies from Chapter 1.

1.7.1 Compiler IR View

At the compiler layer, the mindset shift appears as a change in primary IR artifacts. Operator-first pipelines lower high-level frameworks (Torch, ONNX) into op sequences early; each op lowers independently to LLVM, Triton, or CUDA C++ with late, local fusion peephole passes. Dataflow-first pipelines retain a *lifetime graph* on tensors until hardware tier assignment completes: SSA values carry storage class annotations (`reg`, `shared`, `global_stream`) and fusion legality is checked on inter-value edges. GPU backends emit kernels per fused region; CPU backends emit blocked loops; XDNA backends emit ADF graphs with DMA edges. The IR difference is why "drop in FlashAttention" is not equivalent to "run the fusion pass pipeline": the former swaps an op implementation; the latter rewrites the graph topology.

1.7.2 Failure Modes When Mixing Mindsets

Hybrid stacks—dataflow planning in a planner service but operator-first codegen in backends—produce characteristic failures. **Planner drift:** fusion regions computed on a meta-graph diverge from what the backend actually emits, reintroducing hidden exports. **Partial graph capture:** CUDA Graphs record a subgraph that still contains HBM ping-pong inside recorded nodes. **False residency:** shared-memory buffers declared fused but spilled registers silently increase global traffic. **NPU illegalize late:** graphs accepted on GPU fail XDNA buffer allocation in production edge builds. Guard against hybrids with single-source dataflow IR and fail-fast legality checks per backend, the YiRage default posture.

1.8 Contrasting Schedules on One Decoder Layer

Consider a single transformer decoder layer at batch-1 as the book's running microscope. **Operator-first execution** issues separate kernels for RMSNorm, QKV linear, rotary embedding, attention, output projection, second RMSNorm, gate/up/down MLP projections, activation, and residual adds—each reading inputs from HBM and writing outputs back. Even if every kernel is "optimized," the layer may execute $\sim$10–20 global export/import cycles per token per layer, multiplied by host launches documented in TaxBreak. Profiler heatmaps look healthy inside each kernel; end-to-end latency does not move.

Dataflow-first execution partitions the same mathematics into two or three pipeline stages with explicit buffer assignments. Stage A might fuse RM-

SNorm+QKV+rotary into shared-memory tiles; Stage B runs attention with KV streaming while keeping query tensors resident; Stage C fuses MLP+norm+residual without global activation ping-pong. Launches drop from ∼10–20 toward 1–3; bytes moved drop because producer–consumer pairs never left the fastest tier. On H100-class GPUs this is how CUDA Graphs and MegaKernels compound: graphs fix dispatch; dataflow fixes bytes.

The contrast is sharper on **CPU**: operator-first eager PyTorch thrashes LLC moving the same activation lines through DRAM for every sub-op. Dataflow-first blocking keeps the MLP epilogue in L2 while weights stream once per layer. On **XDNA**, operator-first schedules may not legalize at all; dataflow-first ADF graphs assign each stage to tile columns with DMA prefetch of weights while SRAM holds activations. The layer microscope therefore generalizes: the mathematical layer is fixed; the *schedule topology* is the optimization variable.

1.8.1 Migration Checklist: Operator Stack to Dataflow IR

Teams porting production decoder stacks should not "rewrite everything" day one. A pragmatic migration applies five checks to each layer group:

1. **Export audit.** List tensors written to global memory between mathematically fused-able ops; each edge is a candidate fusion edge.
2. **Launch audit.** Count kernels per token; if $\gg$100 on dense decode, graph capture or fusion passes are higher ROI than MMA tuning.
3. **Tier budget.** Estimate shared-memory or cache footprint for candidate fusion; split stages when capacity exceeds hardware limits rather than spilling silently.
4. **Attention boundary.** Keep KV-global attention stages explicit unless register/shared budgets prove full inlining safe.
5. **Regression matrix.** Re-run bytes/token and ms/token on GPU, CPU, and NPU SKUs after each fusion tranche—not only the device that authored the kernel.

YiRage encodes these checks as pass gates; manual teams should encode them as CI benchmarks even before automation lands.

1.8.2 Organizational Incentives

Operator-first stacks persist partly because performance reviews reward visible kernel wins—ns saved in cuBLAS, occupancy graphs in Nsight—while dataflow wins appear "distributed" across fewer exports and launches. Staffing mirrors the stack: siloed kernel engineers without a compiler owner cannot hold global residency accountability. Industrial leaders align incentives with bytes/token and kernels/token OKRs tied to product p99 latency, not single-kernel FLOP counters. This organizational point matters for adoption: dataflow-first design is a *process* change as much as a code change.

1.8.3 Glossary Bridge

To align vocabulary across hardware chapters: **operator** means a mathematically named function with its own launch or library entry; **stage** means a fused pipeline region with one dispatch and internal register/shared residency; **dataflow** means the directed acyclic graph of tensor lifetimes and movement edges; **residency** means the storage tier holding a value at a program point; **MegaKernel** means a manually authored stage at or near full-layer scope; **automation path** means compiler passes that derive stages from IR instead of hand CUDA/HIP. These terms are used consistently in Chapters 3–11 and in YiRage pass names in Part VI.

1.8.4 Historical Note: Why Operator-First Won (Temporarily)

Operator-first CUDA culture dominated the 2010s because training workloads were compute-heavy, batches were large, and framework ecosystems competed on op coverage. Inference—especially decode—was treated as "training with batch 1" long after production traffic proved otherwise. FlashAttention, PagedAttention, and graph capture each corrected a slice of the inference gap without overturning the op-list mental model. The dataflow-first resurgence in this book is driven by deployment reality: disaggregated decode fleets, MoE launch inflation, edge NPUs, and cross-hardware SKUs that cannot afford per-device hero kernels. Understanding *why* operator-first habits persist helps teams plan migrations without dismissing past kernel investments—those kernels become *epilogues inside stages*, not the architecture itself.

1.8.5 Practice Problems (Engineering, Not Exercises)

Use these audits on your own decoder deployment:

1. Draw the operator graph for one layer and mark every HBM read/write edge; count edges per token.

2. Propose a two-stage dataflow that removes at least half those edges; estimate shared-memory or cache bytes required.

3. Run TaxBreak-style launch counting or equivalent framework profilers; compare kernels/token before and after fusion.

4. Measure R_{floor} on your GPU SKU at batch-1; if high, prioritize bytes; if low, prioritize launches.

5. If targeting NPU, attempt to legalize the same graph statically; document which edges force DDR spill.

Solutions appear implicitly in Parts V–VI through YiRage pass implementations; the audits themselves are portable across vendors.

Readers building internal playbooks should copy the audits into onboarding docs for new kernel hires—especially those arriving from training-focused CUDA backgrounds—so decode residency constraints are explicit on day one rather than rediscovered via quarter-long latency firefights. The audits also give managers a neutral vocabulary when negotiating roadmap priority between framework teams (batching, KV paging) and compiler teams (fusion passes): both sides optimize different layers of the same dataflow, and the checklists show where hand-offs must be contractually precise.

1.9 Dataflow mindset benchmarking methodology

Benchmarking the mindset shift must measure *schedule quality and export topology*, not isolated kernel FLOPs (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024b,a; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Build paired decode cells on the same model: operator-first eager baseline versus dataflow-first fused schedule, BS=1, ctx buckets $\{512, 2048, 8192\}$, pinned weights and framework version (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Metric	Mindset comparison meaning
Launches/token	Operator graph vs fused stages
Bytes/token	HBM export count between adjacent ops
R_{floor}	Memory-bandwidth headroom after fusion
Fusion depth	Number of global exports per layer
Triplet legality	Same IR legal on GPU/CPU/NPU

Table 1.2: Dataflow mindset counter worksheet (extends Chapter 1).

Warm-up must include framework graph construction and at least one thousand steady decode steps—single-step microbenches that never cross operator boundaries mislead teams into operator-first wins (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Report launches/token, bytes/token, and ms/token together; a fusion that cuts launches but not bytes failed residency planning (Chen, 2026; Vellaisamy *et al.*, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Example 1.9.1. Mindset gate: fusing three adjacent ops must cut global exports by $\geq$2 and bytes/token by $\geq$10% at ctx=2048 *or* document why iron rules forced a stage split; attach residency diagram (Chen, 2026; Vellaisamy *et al.*, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Export-edge audit. Draw the operator graph for one decoder layer; count HBM read/write edges per token (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Each edge between mathematically fused-able ops is a dataflow candidate; benchmark before/after removing edges, not before/after retuning MMA tiles (Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Triplet regression discipline. Export the same high-level decoder from YiRage IR to CUDA, CPU, and XDNA backends; diff legality failures before celebrating GPU ms/token (Chen and YiRage Contributors, 2026; Chen, 2020; AMD, 2024b,a; Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Operator-first schedules that legalize only on Hopper should fail CI when edge fleets include Ryzen AI or CPU fallback nodes (AMD, 2024b,a; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Organizational metrics. Align team OKRs with bytes/token and launches/token, not single-kernel TFLOPS (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Review PRs with residency diagrams and export-edge deltas—operator-first microbench wins without counter movement are rejected (Chen, 2026; Vellaisamy *et al.*, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Migration benchmark cadence. When porting operator stacks to dataflow IR, benchmark after each fusion tranche: Stage A (norm+QKV), Stage B (attention boundary), Stage C (MLP+residual) (Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Partial migrations that fuse MLP but leave QKV eager often show launch wins without bytes wins—document both (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Cross-stack triage. When decode latency stalls after a fusion merge, sequence: (1) count export edges; (2) diff bytes/token worksheet; (3) diff launches/token; (4) verify triplet legality; (5) only then tune MMA tiles (Chen, 2026; Vellaisamy *et al.*,

2026; Chen and YiRage Contributors, 2026; Davies *et al.*, 2025; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Skipping export audit keeps teams in operator-first thrash (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Documentation contract. Publish mindset benchmark artifacts with every fusion promotion: operator graph, dataflow stage diagram, export-edge count, bytes/token, launches/token at ctx=2048 (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Without published ledgers, field teams cannot tell schedule regressions from framework drift (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Kwon *et al.*, 2023; Chen, 2026; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Release checklist. Before tagging a dataflow schedule for production, require green triplet compile, export-edge reduction versus baseline, and ms/token delta on all three ctx buckets (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024b,a; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Partial green is how fleets inherit operator-first relapses (Chen, 2026; Vellaisamy *et al.*, 2026; Chen and YiRage Contributors, 2026; Davies *et al.*, 2025; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Handoff to Chapter 3. Mindset benchmarks complete when residency diagrams and triplet legality notes are pinned for the hardware constraint chapter (NVIDIA, 2022; AMD, 2024b,a; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; SemiAnalysis, 2024). Missing artifacts force Chapter 3 readers to guess tier budgets (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020,

2026; Vellaisamy *et al.*, 2026; NVIDIA, 2022; AMD, 2024a,b; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; SemiAnalysis, 2024).

Closing note. Treat dataflow benchmarking as a product contract—without nightly export-edge replay artifacts stored in CI, every latency spike reopens the costly operator-first versus dataflow-first debate without evidence (Chen and YiRage Contributors, 2026; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Speculative and multi-model residency. Draft-verify stacks add second weight streams; benchmark acceptance-rate sweeps with separate export-edge ledgers per model (Davies *et al.*, 2025; Kwon *et al.*, 2023; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024b,a; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Operator-first stacks bolt draft models on without revisiting fusion boundaries—dataflow audits must count both models (Chen and YiRage Contributors, 2026; Kwon *et al.*, 2023; Davies *et al.*, 2025; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Long-context stress. Extend mindset benchmarks to ctx=16384 when paging allows; fusion plans that work at 2K often reintroduce exports at 32K when block tables grow (Kwon *et al.*, 2023; Dao *et al.*, 2022; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Fail CI when export-edge count rises with context while launches/token stays flat (Chen, 2026; Kwon *et al.*, 2023; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Efficiency metric. Track *schedule-attributed decode efficiency*: measured tokens/s divided by roofline tokens/s after subtracting framework overhead, with explicit operator-graph baseline in the denominator (Davies *et al.*, 2025; Chen, 2026;

Vellaisamy *et al.*, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Compiler teams own fused-schedule numerator; framework teams own eager baseline—publish both weekly (Chen and YiRage Contributors, 2026; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Regression harness. Check golden mindset scripts into CI: same model, ctx buckets, export-edge assertions, and TaxBreak launches/token nightly (Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Davies *et al.*, 2025; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Submodule or pass changes that increase export edges without documented iron-rule splits fail the gate (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Summary. Dataflow mindset benchmarks are complete when operator and dataflow worksheets agree within published tolerances on all three ctx buckets—anything else is still an integration draft, not yet a shippable production schedule release candidate (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Continuous batching interaction. Fleet batching improves weight reuse but does not fix per-request export edges inside layers (Kwon *et al.*, 2023; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark BS=1 counters separately from fleet goodput—mindset wins must appear in both (Davies *et al.*, 2025; Kwon *et al.*, 2023; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Quantization and residency. INT8 and FP8 paths need explicit scale tensors in benchmark exports so fusion legality checks do not fold away dequant boundaries (Liu *et al.*, 2024; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Kwon *et al.*, 2023; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Operator-first stacks often dequantize to global buffers per op—dataflow plans must keep scales in-register or in shared tiers (Liu *et al.*, 2024; Chen and YiRage Contributors, 2026; Chen, 2026; Vellaisamy *et al.*, 2026; Chen, 2020; Davies *et al.*, 2025; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Observability hooks. Instrument fused stages with stable names derived from dataflow stage ids so profilers map samples back to export edges without re-running the full compiler (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Production builds strip verbose tags but retain stage hashes in release notes (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Practitioner checklist. Before merging any fusion PR, confirm: residency diagram attached, export-edge delta documented, bytes/token and launches/token updated, triplet export compiles, and end-to-end ms/token measured—never kernel-only timers alone (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Skipping end-to-end comparison is how operator-first microbench wins reach production undetected (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

MoE launch inflation. Mixture-of-experts decoders multiply operator boundaries per layer; mindset benchmarks must count launches/token per expert dispatch, not only dense-layer baselines (Liu *et al.*, 2024; Vellaisamy *et al.*, 2026; Davies

et al., 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Kwon *et al.*, 2023; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Dataflow plans that fuse MLP experts but leave routing eager often show flat bytes/token—document routing tax separately (Liu *et al.*, 2024; Chen and YiRage Contributors, 2026; Vellaisamy *et al.*, 2026; Chen, 2026, 2020; Davies *et al.*, 2025; Kwon *et al.*, 2023; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Operator library trap. Vendor GEMM wins inside isolated ops do not prove operator-first schedules are optimal (Davies *et al.*, 2025; Goodfellow *et al.*, 2016; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark paired schedules on identical library versions; the variable is export topology, not MMA kernel choice (Chen, 2026; Vellaisamy *et al.*, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Mechanical sympathy recap. Fregly's mechanical sympathy appears here as *schedule sympathy*: align compute order with tier capacity before tuning FLOPs (Davies *et al.*, 2025; Goodfellow *et al.*, 2016; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Publish mindset worksheets beside every schedule promotion so Chapter 3 hardware constraints have measurable inputs (NVIDIA, 2022; AMD, 2024b,a; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; SemiAnalysis, 2024).

Prefill versus decode mindset. Prefill benefits from compute-heavy GEMMs where operator-first tuning can help; decode at batch-1 is residency-dominated (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Mindset benchmarks for this chapter focus on decode

cells—do not accept prefill-only evidence as proof of dataflow success (Davies *et al.*, 2025; Kwon *et al.*, 2023; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Operator graph visualization gate. Require a figure or table listing every global export between LayerNorm, QKV, attention, and MLP in the baseline; the dataflow PR must show removed edges highlighted (Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Reviewers who cannot see export topology cannot judge dataflow mindset claims (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Goodput linkage. Mindset shifts matter only when ms/token moves with bytes/token and launches/token—publish all three counters on the same dashboard row weekly (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

CI golden pair policy. Store operator-first and dataflow-first decode scripts side by side in CI; any PR that moves bytes/token or launches/token in the wrong direction versus baseline fails before merge (Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Davies *et al.*, 2025; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Golden pairs prevent silent and costly operator-first relapses when teams optimize isolated kernels without updating the fused schedule (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Closing checklist. Ship operator graph, dataflow stage diagram, export-edge table, and triplet legality summary with every schedule promotion in CI—no exceptions (Chen and YiRage Contributors, 2026; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Worked export audit. For a 7B decoder layer at batch-1, operator-first schedules typically materialize RMSNorm output, QKV tensors, attention output, and MLP intermediates to HBM—four to eight export events before residuals (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). A two-stage dataflow removing norm+QKV and MLP+residual exports cuts events by half class; benchmark bytes/token before claiming stage count (Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Hybrid failure benchmarks. Teams mixing planner-side fusion with operator-first backends should benchmark *planner drift*: compare planned fusion regions to emitted kernels weekly (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Hidden exports reappear when codegen ignores lifetime metadata—CI should diff export-edge counts, not only pass logs (Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

CPU and NPU mindset deltas. GPU mindset wins often fail on CPU when fusion depth evicts KV working sets from LLC (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). XDNA schedules may require layer-scale static regions where GPU schedules used three medium

fusions—triplet benchmarks expose the delta early (AMD, 2024b,a; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Training versus inference mindset guard. Training teams rewarded for large-batch GEMM wins often propose decode "optimizations" that increase exports (Davies *et al.*, 2025; Goodfellow *et al.*, 2016; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Decode mindset benchmarks reject changes that improve prefill TFLOPS but raise launches/token at batch-1 (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

FlashAttention composition benchmarks. Replacing attention ops with FlashAttention without replanning exports fixes attention bytes but may leave norm+QKV and MLP ping-pong intact (Dao *et al.*, 2022, 2023; Chen, 2026; Vellaisamy *et al.*, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark full-layer counters after attention swaps—mindset progress requires layer-scope worksheets (Chen, 2026; Dao *et al.*, 2022; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Graph capture without dataflow. CUDA Graphs recording operator-first subgraphs fix launches but freeze HBM ping-pong inside recorded nodes (NVIDIA, 2024; Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark bytes/token alongside graph capture—graphs are orchestration relief, not residency fixes (Chen, 2026; NVIDIA, 2024; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; AMD, 2024a,b; Goodfellow

et al., 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Iron rules in benchmarks. Each benchmark report should state which iron rule justified every global export that remains (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Unexplained exports fail review—dataflow mindset is auditable, not aesthetic (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Automation path benchmarks. YiRage compile failures on illegal residency are preferable to silent slowdowns—track compile reject rates and reasons in CI dashboards (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Rising reject rates often indicate operator-first edits bypassing lifetime IR (Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Fleet A/B discipline. Canary dataflow schedules must compare export-edge counts and bytes/token against operator baseline on identical traffic slices—not only p50 latency (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Tail latency improvements without counter movement suggest queue luck, not mindset wins (Davies *et al.*, 2025; Kwon *et al.*, 2023; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Documentation for operators. Publish mindset migration playbooks with export audits, migration checklist results, and triplet legality summaries in versioned artifact stores (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*,

2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Partners cannot replay schedule regressions when benchmarks stay in private spreadsheets (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Stage sizing benchmarks. When splitting a layer into stages, benchmark shared-memory or cache bytes per stage against hardware caps (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; NVIDIA, 2022; SemiAnalysis, 2024; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023). Stage splits justified only by spills should show bytes/token regression versus a smaller fusion—document the iron-rule violation that forced the split (Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Attention boundary benchmarks. Keep KV-global attention as an explicit stage unless register and shared budgets prove safe inlining (Dao *et al.*, 2022, 2023; Kwon *et al.*, 2023; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark attention-stage bytes separately from MLP fusion—teams often conflate FlashAttention wins with full-layer dataflow wins (Dao *et al.*, 2022; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Residual and norm fusion benchmarks. RMSNorm and residual adds look cheap in operator-first profilers but dominate export counts when every sub-op writes HBM (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark norm+MLP fusion tranches

explicitly—early wins here build organizational trust in dataflow migration (Chen, 2026; Vellaisamy *et al.*, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

PagedAttention interaction benchmarks. Paging fixes KV capacity, not activation residency between ops inside a layer (Kwon *et al.*, 2023; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark paging layouts with identical fusion plans—mindset metrics must not confound paging block size changes with export removal (Kwon *et al.*, 2023; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Multi-GPU mindset benchmarks. Tensor-parallel decode splits heads but should not reintroduce operator-first exports between shards (Liu *et al.*, 2024; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Kwon *et al.*, 2023; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark per-rank export-edge counts; balanced fleet averages hide straggler ranks with operator-style shard boundaries (Chen, 2026; Vellaisamy *et al.*, 2026; Chen and YiRage Contributors, 2026; Liu *et al.*, 2024; Chen, 2020; Davies *et al.*, 2025; Kwon *et al.*, 2023; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Release train gates. Block production promotion when operator baseline and dataflow candidate worksheets diverge without signed iron-rule exceptions (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Partial promotion is how fleets run mixed schedules behind one load balancer—bimodal latency follows (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Kwon *et al.*, 2023; Chen, 2026; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Final summary metric. Define *export reduction ratio* = (baseline export edges − fused export edges) / baseline export edges; require ≥0.4 for layer-scope promotions unless documented and reviewed attention-boundary exceptions apply (Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Pair with bytes/token and launches/token gates—export ratio alone is insufficient but catches fake fusion (Chen, 2026; Vellaisamy *et al.*, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Onboarding benchmarks. New kernel hires from training backgrounds should run export audits on day one before proposing MMA tile changes (Davies *et al.*, 2025; Goodfellow *et al.*, 2016; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Onboarding docs must include paired operator and dataflow worksheets—culture follows metrics (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Vendor library A/B policy. When swapping cuBLAS or hipBLAS versions, re-run mindset worksheets—library speedups inside ops do not validate operator-first schedules (Davies *et al.*, 2025; Goodfellow *et al.*, 2016; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Hold fusion topology constant across library bumps (Chen, 2026; Vellaisamy *et al.*, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Framework plugin boundaries. Benchmark in-step counters separately from framework queue time—mindset wins must appear inside the decode box defined in Chapter 1 (Kwon *et al.*, 2023; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024;

NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Plugin integrations that add host syncs can erase fusion wins unless worksheets isolate framework tax (Davies *et al.*, 2025; Kwon *et al.*, 2023; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Regression bisect protocol. When bytes/token regresses, bisect fusion passes before bisecting MMA tiles—export-edge diffs localize faster than ns-level kernel diffs (Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). When launches/token regresses, inspect graph capture and framework dispatch before rewriting dataflow stages (Vellaisamy *et al.*, 2026; NVIDIA, 2024; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Closing synthesis. Dataflow mindset benchmarking is complete when export reduction ratio, bytes/token, launches/token, and ms/token align on three ctx buckets with triplet legality pinned—ship the worksheets publicly or keep the schedule in lab only (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Rotary and QKV fusion benchmarks. Rotary embedding between QKV and attention is a frequent hidden export in operator-first stacks (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Benchmark norm+QKV+rotary fusion as Stage A; skipping rotary in export audits understates operator-first tax (Chen, 2026; Vellaisamy *et al.*, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

SwiGLU MLP fusion benchmarks. Gate, up, and down projections triple export events in operator-first MLP chains (Vellaisamy *et al.*, 2026; Chen, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Dataflow Stage C should benchmark SwiGLU as one residency region—partial down-proj fusion without gate/up is a common false win (Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Cross-chapter worksheet handoff. Attach completed mindset worksheets to Chapter 3 hardware constraint reviews so tier budgets are evidence-backed, not assumed (NVIDIA, 2022; AMD, 2024b,a; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; SemiAnalysis, 2024). Chapter 3 readers should not re-derive export counts that Chapter 2 already measured (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; NVIDIA, 2022; AMD, 2024a,b; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; SemiAnalysis, 2024).

Nightly CI cadence. Run mindset golden pairs nightly on pinned GPU CI runners with fixed seeds; CPU and NPU triplet legs weekly when hardware pools allow (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Flaky tests that skip export assertions teach teams to ignore critical dataflow mindset gates in every release (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Executive readout template. One-page promotion summaries must show operator versus dataflow counter deltas side by side—executives approve schedules, not kernel names (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*,

2023; NVIDIA, 2022; SemiAnalysis, 2024). Hide counter regressions behind p50 latency charts and fleets rediscover operator-first habits (Davies *et al.*, 2025; Kwon *et al.*, 2023; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Postmortem discipline. Decode incidents should replay export-edge counts from canary worksheets before blaming drivers or hardware (Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Missing worksheets force multi-week debates that dataflow benchmarks are meant to prevent (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Benchmark artifact schema. Store JSON lines per decode step with schedule id, export-edge count, bytes estimate, launches count, and ctx length (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Aggregates belong in dashboards; per-step JSON enables postmortems (Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Kwon *et al.*, 2023; Chen, 2026; Vellaisamy *et al.*, 2026; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Final gate sentence. Do not merge mindset changes without green export reduction ratio and triplet compile on the same pinned artifact hash (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Anything less is costly operator-first drift wearing dataflow labels (Chen, 2026; Vellaisamy *et al.*, 2026; Davies *et al.*, 2025; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Worksheet archive policy. Version mindset worksheets in the same artifact store as compiler fingerprints so bisect can compare export topology across releases (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Kwon *et al.*, 2023; Liu *et al.*, 2024; NVIDIA, 2024; Dao *et al.*, 2022; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Schedule sympathy mantra. Profile exports first, fuse second, tune FLOPs third, re-profile on every ctx bucket change (Davies *et al.*, 2025; Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024). Post the schedule sympathy mantra beside every mindset worksheet template in the team wiki (Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Chen, 2026; Vellaisamy *et al.*, 2026; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

Acceptance criteria recap. Green mindset promotion on all three ctx buckets requires export reduction ratio $\geq$0.4, bytes/token down, launches/token down or flat, ms/token down, and triplet legality on pinned hashes (Vellaisamy *et al.*, 2026; Chen, 2026; Chen and YiRage Contributors, 2026; Chen, 2020; Davies *et al.*, 2025; Dao *et al.*, 2022; Liu *et al.*, 2024; Kwon *et al.*, 2023; NVIDIA, 2024; AMD, 2024a,b; Goodfellow *et al.*, 2016; Dao *et al.*, 2023; NVIDIA, 2022; SemiAnalysis, 2024).

1.10 Key Takeaways

- **Operator-first thinking is the default trap.** Listing a decoder as LayerNorm, QKV, attention, MLP, and residuals and tuning each kernel in isolation stitches them with separate launches, so the failure is structural rather than per-kernel (Dao *et al.*, 2023; Vellaisamy *et al.*, 2026). No amount of single-operator tuning removes the launch-and-spill tax that the operator boundaries themselves create (Chen, 2026).
- **Residency, not FLOPs, is the root cause.** Activations that could live in registers or shared memory across the QKV→attention→MLP chain are instead materialized to HBM at every operator boundary, which is the real source of decode latency (Chen, 2026; Dao *et al.*, 2022). The dataflow mindset

names the schedule and tensor lifetimes before operators so those activations stay on-chip (Chen and YiRage Contributors, 2026).

- **The iron rules assign each tier its job.** Registers hold ephemeral scalars and narrow vectors, on-chip SRAM holds fusion regions up to its capacity, and a spill is the signal the split is wrong—these are checkable invariants, not guidelines (Chen and YiRage Contributors, 2026; Goodfellow *et al.*, 2016). Encoding them as metadata is what lets a compiler reject an illegal schedule before codegen (Chen, 2020).

- **Cross-hardware is a design input, not an appendix.** A GPU-tuned schedule silently fails on an edge NPU's static fabric and underperforms on a co-located CPU node, so the mindset must be multi-target from the start (AMD, 2024b,a). The same dataflow plan yields different bytes/token per class, which only a cross-hardware comparison reveals (Vellaisamy *et al.*, 2026).

- **The MegaKernel is the mindset materialized.** Dispatching an entire decoder layer with activations resident across QKV, attention, MLP, norms, and residuals is the antithesis of the operator graph, and YiRage automates it by planning lifetimes globally from one dataflow IR (Chen and YiRage Contributors, 2026; Chen, 2026). The shift is from optimizing kernels to optimizing the schedule that ties them together (Davies *et al.*, 2025).

1.11 Conclusion

The operator-driven and dataflow-driven mindsets are not two styles of the same craft; they are different theories of what a decoder layer *is*—a list of kernels to tune versus a schedule of residency to plan (Vellaisamy *et al.*, 2026; Chen, 2026). The dataflow mindset is the one this book commits to, because at batch-1 decode the operator boundaries themselves are the cost, and only a residency-first schedule—materialized as a MegaKernel and enforced by YiRage's iron rules—removes them (Chen and YiRage Contributors, 2026; Dao *et al.*, 2022). That mindset is empty, however, without the hardware facts it must plan against: Chapter **??** supplies the concrete memory hierarchies, parallelism grains, and scheduling models of GPUs, CPUs, and NPUs that every later compiler pass encodes (NVIDIA, 2022; AMD, 2024a).

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